Course #: DS6050: ML III: Deep Learning
Syllabus: Spring 2026

Course Information

Course Start Date–End Date: January 12, 2026 – May 17, 2026
Meeting Days & Times: Monday, Wednesday, and Fridays from 10-10:50 (Section 1) or 11-11:50 (Section 2)

Instructor Information

Dr. Tom Hartvigsen
Assistant Professor, School of Data Science
Email: hartvigsen@virginia.edu
Website: https://www.tomhartvigsen.com
Office Hours: Thursdays 4:00-5:00
Office Location: SDS 339

TA Information

- **Eric Onyame** (PhD Student, School of Data Science)
  Email: reh6ed@virginia.edu
  Office Hours: 2-3PM, Tuesdays, SDS 300
- **Eirik Drage Steen** (PhD Student, School of Data Science)
  Email: zxc6hs@virginia.edu
  Office Hours: 1PM, Wednesdays,
  https://us05web.zoom.us/j/85746649915?pwd=HYZPoxarQu9mhEnE32graJBQqn2pEa.1

Course Overview

This is a graduate-level course on deep learning, aiming to cover fundamental concepts for deep neural networks and their applications. Topics include multi-layer perceptron, backpropagation, convolutional neural networks, recurrent neural networks, generative networks, deep reinforcement learning, interpretable deep learning, etc. The course will focus on hands-on learning activities using open-source Python libraries such as NumPy, scikit-learn, and PyTorch. Students are required to have sufficient computational background to complete several substantive programming assignments.

Learning Outcomes

By the end of the semester, you will be able to:

1. Create an end-to-end deep learning project at scale using open-source libraries such as NumPy, scikit-learn, and PyTorch.
2. Formulate various supervised, unsupervised, and reinforcement learning models.
3. Apply practical skill sets on designing, deploying, and analyzing deep network architectures on complex real-world problems.
4. Articulate concepts, algorithms, and tools to build intelligent systems.
5. Analyze advanced approaches currently pursued by the machine learning community.

Semester Schedule

Always refer to the Canvas calendar tool for accurate dates and deadlines

Module	Date	Topic	Note
0. Introduction & ML recap	01/12	Welcome & Course Overview	
	01/14	Math Foundations of Deep Learning	
	01/16	Introduction to Machine Cognition	
1. Artificial Neural Networks	01/19	No Class --- Holiday	
	01/21	Multi-Layer Perceptrons & Deep Neural Networks	
	01/23	Optimizing Deep Neural Networks (Gradient Descent & Backpropagation)	A1 Assigned
2. Convolutional Neural Networks	01/26	Convolutional Neural Networks	
	01/28	Basic Building Blocks of CNNs	
	01/30	In-Class Exercise	
3. Training CNNs 1	02/02	Data distribution, Regularization	
	02/04	Data processing, Augmentation	A1 Due
	02/06	In-Class Exercise	A2 Assigned
4. Training CNNs 2	02/09	Optimizing CNNs	

	02/11	Loss Functions & Other Tips	
	02/13	In-Class Exercise	
5. CNN Architectures	02/16	Early CNN Architectures	
	02/18	Modern CNN Architectures	
	02/20	In-Class Exercise	
6. Project Planning	02/23	Project Overview + Q&A	
	02/25	No Class	
	02/27	Proposal Workday	A2 Due
No Class --- Spring Recess	03/02	No Class --- Spring Recess	
	03/04		
	03/06		
7. Sequence Modeling	03/09	Sequence Data & Word Embeddings	
	03/11	RNN Architectures & Attention Mechanisms	
	03/13	In-Class Exercise	Project Proposal Due
8.0 Large Language Models (Part 1)	03/16	Language modeling & data collection	
	03/18	Transformer Architectures	A3 Assigned
	03/20	In-Class Project Work Day	
8.1 Large Language Models (Part 2)	03/23	Modern Transformer Architectures	
	03/25	Pre-training Large Language Models	
	03/27	In-Class Project Work Day	Project Checkpoint Due
8.2 Large Language Models (Part 3)	03/30	Finetuning Language Models	
	04/01	Post-training Language Models	
	04/03	In-Class Project Work Day	A3 Due
9. Generative Networks	04/06	Encoder-Decoder Architectures	
	04/08	Diffusion Models	
	04/10	In-Class Project Work Day	
10. Advanced Topics	04/13	Deep Reinforcement Learning	
	04/15	Multi-modal Deep Learning	
	04/17	In-Class Project Work Day	Project Slides Due
11. Final Presentations	04/20	Final Presentations	
	04/22	Final Presentations	
	04/24	Final Presentations	
14. Recap & Summary	04/27	Recap & Summary	Project Report Due

Group Project

The course project will be undertaken as a group project and will be a total of 350 points (or 35% of the final grade) as broken down below:

1. Project proposal & literature review (100 pts)
2. Project checkpoint (data pipeline and model prototypes) (50 pts)
3. Project presentation (100 pts)
4. Project report (100 pts)

Details of the groups are:

- The recommended group size is 3 to 5. Individuals may be able to work by themselves with good reason. Expectations will not be adjusted due to the smaller group size. Groups of more than three are strongly discouraged but can be allowed under extreme circumstances. Expectations for groups of four are higher than those for groups of three.
- In general, all group members will receive the same grade for graded assignments. However, group members will evaluate their peers, and any student who appears to not be contributing may be penalized.
- Each group will be responsible for assigning tasks to its group members.
- You are expected to work as a member of your group in this course and cooperate with your colleagues. There will be multiple working sessions and checkpoints to help your team to stay on track with the project. Cooperation means attending group meetings, completing your assignments properly and on time, letting your group know if you will be out of town, responding to emails from your group, and so on. If there is a lack of cooperation by any group member, it must be brought to the attention of the instructor as soon as it happens. If the lack of cooperation is serious, the offending group member's semester grade will be penalized

What to Expect in This Residential Course

Students are expected to actively participate in class discussions by asking questions and contributing to the discussion.

Students are expected to come to class prepared to participate in each week's activity. Learners are expected to complete the homework prior to class.

Students may work together on homework assignments but each student's answers must be in their own words.

Zoom opportunities may be offered throughout the semester to allow for students who are traveling or home with illness to still participate in class discussions and activities. As a

Residential program, students are expected to attend class in-person unless occasional and specific circumstances require a virtual option.

Learning Resources

Textbook/Readings

- Goodfellow, Bengio, and Courville, ***Deep Learning*** (Cambridge, MA: MIT Press, 2016). *Free online copy of this book is also available at*: http://www.deeplearningbook.org/
- Antiga, Stevens, Viehmann, ***Deep Learning with PyTorch*** (Manning Publications, 2020). *Online available at*: https://www.oreilly.com/library/view/deep-learning-with/9781617295263/

* The field of deep learning is rapidly evolving. To keep up with the newest trends, the course will also be based largely on conference publications and online materials.

Grading

Programming Assignments	Programming assignments will be implemented in Jupyter Notebooks and provide hands-on experience writing/modifying Python/PyTorch code while working with various datasets.
Module Quizzes	At the end of each module, a module quiz will assess student knowledge and application of topics covered in reading assignments and modules.
Group Project	The group project is a large component of the course; it includes students working in groups of three to work on a real-world deep learning challenge. Groups are free to choose their own topic of interest from sponsored projects, public data science competitions (e.g., Kaggle), or other areas of deep learning research and/or applications. The project consists of four major milestones: topic selection/proposal, progress checkpoint, presentation, and final report.

Assignments will be graded according to the following breakdown:

Assignment	**Total Points**	**Percentage**	**Modules**
3 Programming Assignments	450 (3 x 150) pts	45%	M2, 3-6, 9
10 Quizzes	200 (10 x 20) pts	20%	Pick 10 from M1-11
1 Group Project	350 (1 x 350) pts	35%	M1-11

The final letter grade will be determined by the following scale:

Grade	**From**	**To**

A+	970	1,000
A	930	969
A-	900	929
B+	870	899
B	830	869
B-	800	829
C+	770	799
C	730	769
C-	700	729
D+	670	699
D	630	669
D-	600	629
F	0	599

Programming Assignments

There will be three programming assignments. Each assignment will be graded for correct implementation, results, a thorough discussion of the results, and good organization, grammar, and spelling. This is a programming course and doing the programming assignments is absolutely critical (and accounts for 45% of your grade). If the results you see when running your program do not match the results that the TA reports for your program, you should talk to the TA promptly. However, in all but the rarest of cases, you should be prepared to understand that the definitive test of your code is that made by the TA, and what you are seeking in talking to the TA is an understanding of how to avoid losing points over similar discrepancies in the future. To put this another way, you should contact the TA, but if you understand why there is a difference, you should not assume that your assignment will be re-graded.

Expectations

The expectation is that all assignments, live coding, quizzes, discussions, notebooks, and the final project will be submitted on time, by the due date and time/time zone, indicated in Canvas. Submitting your work on time ensures you learn pre-requisite skills and concepts and that you are prepared to learn the next set of skills.

Late Work Policy

Falling behind can have serious consequences. Assignments and learning activities prepare you to complete subsequent assignments and continual lateness will impact your workload over the term and may prevent you from catching up. If you do fall behind, you should always complete the earlier assignments first.

All work must be completed by the end of the semester regardless of extensions/grace period. Work submitted after the last day of the semester will not be considered in the computation of the final grade.

Extenuating Circumstances

Students are expected to communicate with faculty as soon as possible regarding extenuating circumstances and how their participation in the course may be affected, including attendance, assignment submissions, and extension requests.

If you need help, are falling behind and struggling to catch up, or if extenuating circumstances arise, reach out for help from the TAs or the instructor.

Accessing Grades and Feedback in Canvas

- Assignments will be graded within a week of their due date. See comments in Gradescope (need to select > to see comments from instructor).
- If you offer feedback in advance of a submission, let students know dates for submitting to receive and apply the feedback.

Class Attendance

Students are expected to attend all class sessions. Instructors establish attendance and participation requirements for each of their courses. Class requirements, regardless of delivery mode, are not waived due to a student's absence from class. Instructors will require students to make up any missed coursework and may deny credit to any student whose absences are excessive. Instructors must keep an attendance record for each student enrolled in the course to document attendance and participation in the class.

Course Policies

Communication

Discussions will be held using Discord. Please use the following link to join the class discord: https://discord.gg/Pq2q4q8M4p. You must set your username to be "Your Name (computing_id)". For example, Tom Hartvigsen's username would be "Tom Hartvigsen (zcy9jk)". This discord server will be your main way to communicate with peers, TAs, and the instructor for all topics, including specific assignments, group project concerns, and individual messages. Students are encouraged to check the discord server daily for updates or other correspondences. Please avoid emailing the instructor when possible, as responses may be further delayed, but use your judgement. Feel free to communicate as needed per the medium of your choice, especially as issues arise. The sooner you inform me of any problem (personal or academic) that may affect your attendance or performance, the better the chance we have of solving it together.

Recording of Class Sessions

Class sessions for this course will not be recorded. Recording during the lectures is not permitted. This includes video and audio.

Canvas Email

Canvas email will be used for course-wide announcements. You are encouraged to NOT use the canvas email to directly connect with the instructor.

Academic Integrity: Collaboration and Cheating

Assignments are graded individually. But you are allowed and encouraged to work together collaboratively. However, the line between collaboration and cheating can get a bit fuzzy. In general, the difference between collaboration and cheating comes down to intent. Cheating is trying to circumvent the learning process. Collaboration is trying to help yourself and your classmates learn the material more deeply. Use your own sense of right and wrong, but to help clarify the difference here are **examples of cheating**:

1. Directly copying someone else's text word for word, or copying text from ChatGPT
2. Sharing/showing code for the purpose of circumventing the learning process (for example, letting someone copy code because they are running up against the deadline)
3. Asking for help without doing anything to try to solve the problem first; asking someone to do the work for you
4. Making your homework freely available on GitHub or another website
5. Sharing answers to reading quizzes

Here are things that are okay:

1. Two people with the same code is okay as long as one person didn't copy-and-paste it from another person
2. Sharing/showing individual lines of code for the purpose of teaching/explaining or helping someone understand the material
3. Debugging together (this is only possible if both people have already written their own code, otherwise there's nothing to debug)
4. Sharing strategies, external texts, blogs, and other resources for completing problems on the lab assignments

If you work with someone else, please just name them and describe how you collaborated in 1-2 sentences.

There are many ways in which I am limited in my ability to enforce these rules, but I ask you to promise on your honor to not to share code or quiz answers. Cheating means that you do not give yourself the opportunity to master the skills to start working with data in Python. Why rob yourself? If you are stressed out about the intensity of the course, please message me and we can work together to get you back on track.

SDS Guidelines on AI Tools and Assistance

The use of generative AI tools and foundation models, (i.e. ChatGPT GPT, DALL-E, Stable Diffusion, Midjourney, GitHub Copilot, and similar tools) is permitted with the following activities in accordance with the stated guidelines at no penalty:

- Brainstorming and refining your ideas;
- Fine tuning your research questions;
- Finding information on your topic;
- Drafting an outline to organize your thoughts; and
- Checking grammar and style.

Students are responsible for

- Acknowledging that large language models tend to produce incorrect facts and fake citations.
- Acknowledging that code generation models may produce inaccurate outputs.
- Acknowledging that image generation models can occasionally come up with highly offensive products.
- Taking responsibility for any inaccurate, biased, offensive, or otherwise unethical content submitted, regardless of the origin (i.e. student-generated or from a foundation model).
- Properly citing the contribution of the foundation model or other AI tools in submitted material.
- The entirety of any information they submit, based on an AI query or AI assistance.

The use of generative AI tools is NOT permitted for the following activities:

- Impersonating students in classroom contexts, such as by using the tool to compose discussion board prompts or content entered into a Zoom chat.
- Completing group work assigned to a student, unless it is mutually agreed upon that they may utilize the tool.
- Writing a draft of a writing assignment.
- Writing entire sentences, paragraphs or papers to complete class assignments.

Students may be penalized for

- Using a foundation model without including an acknowledgement.
- Improperly citing the use of work by other human beings or the submission of work by other human beings as that of the student.
- Violating intellectual property laws.
- Submitting materials containing misinformation or unethical content.

The usage of AI tools must be properly cited to stay within university policies on academic honesty. Failure to adhere to these guidelines will result in a failing grade on the assignment or exam (a zero) and may be an honor code violation depending on the context (to be determined at the instructor's discretion). Having said all these disclaimers, the use of foundation models is encouraged, as it may make it possible for you to submit assignments with higher quality, in less time.

School of Data Science Support and Policies

Office of Student Support

The Office of Student Support is here to support your academic journey toward success. Their office provides resources to help you be successful and engaged. If you need support resources related to student success or personal well-being, please reach out to the Office of Student Support directly at: SDS_OSS@virginia.edu.

Kylen Baskerville, your Graduate Program Manager, is your go-to resource for academic advising and providing personal support when you need it. Connect with Kylen for questions around course enrollment and choosing your electives, getting approval for an internship, or if you're feeling behind in your courses or having trouble balancing your academic and personal life.

The Data Science Student Portal has resources and information about career, academic support, engagement opportunities, funding, and more.

Career Services

The School of Data Science Career Services Team provides a wealth of opportunities for you to learn, connect, and grow. These offerings are adapted to the needs of the cohort or the year and may take different forms as needs change.

Complete your profile and career interests, explore the Data Analytics Resource Card or make an appointment with Career Services at the School of Data Science by using your UVA Email to Login to Handshake.

Graduate Record
Visit the School of Data Science Graduate Record for policies and information about academic regulations, academic standing, financial assistance, and grades.

University Policies

Academic Integrity and University of Virginia Honor System
The School of Data Science relies upon and cherishes its community of trust. We firmly endorse, uphold, and embrace the University's Honor principle that students will not lie, cheat, or steal, nor shall they tolerate those who do. We recognize that even one honor infraction can destroy an exemplary reputation that has taken years to build. Acting in a manner consistent with the principles of honor will benefit every member of the community both while enrolled in the School of Data Science and in the future. Students are expected to be familiar with the university honor code, including the section on academic fraud.

All work should be pledged in the spirit of the Honor System of the University of Virginia. The instructor will indicate which assignments and activities are to be done individually and which permit collaboration. Students who submit exams electronically acknowledge the Honor Pledge by agreeing to the following statement:

"On my honor, I have neither given nor received aid on this examination, nor did I have prior knowledge of its contents."

For more information, please visit UVA Honor Committee.

Course Evaluations: Student feedback is critical to the school, the instructor, and future students. Students are expected to complete anonymous and confidential course evaluations in a timely manner for each course at the end of each term.

Discrimination/Harassment/Retaliation
UVA prohibits discrimination and harassment based on age, color, disability, family medical or genetic information, gender identity or expression, marital status, military status (which includes active duty service members, reserve service members, and dependents), national or ethnic origin, political affiliation, pregnancy (including childbirth and related conditions), race, religion, sex, sexual orientation, veteran status. UVA policy also prohibits retaliation. All faculty and TAs are also responsible employees for disclosures or reports of potential discrimination, harassment, and retaliation.

Disability and Pregnancy Accommodations
If you anticipate or experience any barriers to learning in this course, please discuss your concerns with me. If you have a disability, or think you may have a disability, contact the Student Disability Access Center ("SDAC") to request reasonable accommodation(s) for this course. If you have accommodations through SDAC, send me your Faculty Notification Letter as soon as possible and meet with me so we can develop an implementation plan together.

Students may be entitled to reasonable accommodations for pregnancy, childbirth, or related medical issues. Please contact SDAC for additional information. Pregnant and parenting students are encouraged to contact SDAC or EOCR to discuss plans and ensure ongoing access to their academic courses and program. Information for pregnant and parenting students is also available on EOCR's Pregnancy and Parenting Resources webpage.

Religious Academic Accommodations
UVA also provides reasonable accommodations when a student's sincerely held religious beliefs or observances conflict with academic requirements. Students who wish to request an academic accommodation for a religious observance should submit their request to me by email as far in advance as possible.

If you have questions or concerns about your request, you may contact EOCR at UVAEOCR@virginia.edu or (434) 924-3200 or visit their Religious Accommodations webpage for additional information. Please note that receiving an accommodation does not relieve you of your responsibility to complete any coursework you miss as a result of the accommodation.

Reporting an Incident
Just Report It is the University's online system for reporting:

- Sexual and Gender-Based Harassment and Violence
- Discrimination, Harassment, and Retaliation
- Hazing
- Clery Act Compliance (by CSAs)
- Interference with Speech Rights
- Youth Protection
- Preventing & Addressing Threats or Acts of Violence

You may access Confidential Resources if you wish to discuss a concern or incident without reporting to the University. Information you share with Confidential Resources will not be disclosed to University officials or any other person except in extremely limited circumstances. For more information on reporting an incident, visit this website.

Student Mental Health and Wellbeing

The University of Virginia is committed to advancing the mental health and wellbeing of its students, while acknowledging that a variety of issues directly impact your academic performance. Residential MSDS students may access the School of Data Science Embedded Psychotherapist Beth Holt Wright, LCSW by:

- Scheduling online through the Healthy Hoos Portal or
- Emailing mdj7wf@virginia.edu or
- Calling CAPS at 434-243-5150. Notify the receptionist that you are enrolled in the School of Data Science. If you need expedited / urgent support, consider a walk-in appointment at the main CAPS location on the 4th floor of the Student Health and Wellness Center.

If you or someone you know is feeling overwhelmed, depressed, and/or in need of support, contact the CAPS Care Managers at CAPSCareMgrs@virginia.edu.

For help finding a community therapist, visit the Community Referrals page through CAPS.

Additional Resources

The School of Data Science Office of Student Support can help find resources for students experiencing Emergency Needs. Please contact them at SDS_OSS@virginia.edu.